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Assessing the Galea Headset on a Benchmark of Deep Learning Architectures and Pre-processing Pipelines for EEG Motor-Imagery Task Classification

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I hereby declare that this dissertation is all my own work, except as indicated in the text:

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Date 27th /August / 2026

Abstract

Motor imagery BCI's decode imagined movement from electroencephalography, despite this most published classifiers are evaluated on dense research grade montages under conditions that do not typically transfer to consumer hardware.

This dissertation aims to benchmark three architectures: EEGNet, EEG-Conformer and the novel GRFNet (Galea ROCKET-Fusion Network), primarily on the sparse four channel montage of the OpenBCI Galea headset, but also the native dataset channels, under a controlled, leakage safe evaluation framework.

A controlled experimental grid crosses three architectures with four preprocessing pipelines across five seeds. Using subject specific evaluation on BCI competition IV 2a and 2b with 180 cells overall. Backing this up is non parametric statistical testing utilising Holm-Bonferroni correction, confidence intervals and rank-biserial effect sizes. A cross subject-hardware transfer study tests all trained models and preprocessing pipelines on three participants recorded with the Galea headset.

On the sparse montage, EEGNet leads with 0.599 on four class accuracy, while GRFNet matches EEG-Conformer at 0.566 (GRFNet containing 3655 parameters against Conformers 789,572). Heavier preprocessing displays an advantage on IV-2a whereas lighter preprocessing helps IV-2b. Restricting from 22 channels to four costsist from pipeline to pipeline, and whether different evaluation methods (LOSO and subject-specificThe main conclusions, implications, any challenges encountered and avenues for

future research accuracy, which presents as significant in nine of twelve cells, performance still remains above chance despite this. Cross subject-hardware transfer to galea presents chance level accuracy across every single cell - signal quality diagnostics trace this to electrode placement, material and conductivity.

The key contribution establishes that dry electrode pillars on the Galea headset impose a signal quality issue based on inadequate electrode contact, conductivity and high impedance. The benchmark further demonstrates another finding that under controlled standardised conditions across preprocessing pipelines, compact decoders are able to match or exceed transformer scale models which supports the hypothesis of parameter efficient designs as the most practical path for resource constrained environments.

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Chapter 1

Introduction

1.1 Motivation

There is increasing evidence that addresses various models reporting performance gains which claims advantages that may not be completely down to the models architectural development, but rather the specific preprocessing decisions that are made during the development of the model, decisions which may not have been reported in the methodology reports due to it pertaining to the sequence of application or simply that the preprocessing techniques are favourable to the architecture.

The value of a Brain-Computer Interface is that its reliability can be extrapolated with real world applications, if a decoder can't generalise across preprocessing pipelines, subjects or evaluation methods the practical use is minimal, costly and unreliable. Given that this is common throughout the field, it warrants this project to develop a controlled benchmark that varies architectures and pipelines on a cheaper, consumer grade headset comparable to a published dataset to determine differences

1.2 Aims and Objectives

The aim of this project is to benchmark the extent in which reported accuracies are attributable to preprocessing pipeline choices or whether they are due to actual architectural advances. As well as this, it would be key to establish whether any architectural differences

persist across pipelines, evaluation methods and EEG montages. The three architectures that will be used increase in complexity. Firstly, EEGNet as a Convolutional Neural Network baseline model developed by [Lawhern et al., 2018]. Secondly, a convolutional transformer by [Song et al., 2023] and finally a *novel* Rocket Fusion model which combines ROCKET/MiniRocket developed by [Dempster et al., 2019, Dempster et al., 2021] with a *geometry seeded spatial front end* and *fusion pooling*

1. **Reproducible data pipelines.** Implement end-to-end data pipelines for BCI IV-2a and IV-2b with four preprocessing configurations and two EEG montages. *Success:* all four pipelines run on every grid cell.

2. **Accurate architectural implementation** Implement EEGNet, EEG-Conformer and a GRFNet behind a common interface. *Success:* each model reproduces it's published baseline accuracy within a defined margin of error.

3. **Experimental grid control** Design and execute a $3 \times 4 \times 1$ grid of architectures pipelines and evaluation methods respectively, with five random seeds per cell on the Galea montage. Reporting four key metrics . *Success:* every cell is populated with complete documented metric reports.

4. **statistical analysis** Evaluate the grid using paired tests across seeds, confidence intervals, effect sizes and multiple comparison correction. *Success:* Analysis outputs sufficient results that are discussed.

1.3 Tasks

1. **Virtual environment and dependency setup** Create a virtual environment with a 'requirements.txt' file containing specified versions for Python 3.11, Pytorch 2.12 with CUDA 13, MNE-Python, and other relevant python libraries. Initialise the Git repository with relevant license and environment pushed with a test to identify the installed versions.

2. **Dataset Download** Download the BCI Competition IV-2a and BCI IV-2b from the BBCI archive.

3. **Data loader** Implement a data loader module that provides trial level epochs with consistent shape, labelling and metadata across both datasets. Unit tests must be developed to provide distribution and trial counts compared to published figures.

4. **Preprocessing configuration** Implement all four preprocessing pipelines that are loaded through configuration files.

5. **Determinism tests** Add tests verifying each preprocessing configuration produces identical output across seeds. Stable output within a margin of error should be produced when tested across different seeds.

6. **EEGNet development** Implement the EEGNet architecture and validate accuracy against published baseline on BCI IV-2a within a margin of error. EEGNet is well documented and has maintained a reference implementation.

7. **EEG Conformer development** Implement the EEG conformer architecture, preserving the convolutional patch embedding and transformer encoder structure described by [Song et al., 2023]. Validate the accuracy against the published BCI IV-2a result and document any alterations of the original code.

8. **GRFNet development** Implement the Galea ROCKET fusion network. This is a novel architecture for this project so all design decisions must be recorded with rationale in the methodology. The model must be stable enough to reliably train on the BCI IV-2a and reach the EEGNet baseline accuracy. This is the task with the least amount of clarity for deliverable success.

9. **A unified training interface** Collate all three architectures behind a single training interface with a matched optimiser, batch size, epochs, early stopping criteria and data augmentation policy so that the benchmark compares architectures rather than training recipes. Each model must be runnable from one CLI entry point with selection functions.

10. **Primary and secondary grid execution** Run $3 \times 4 \times 1$ experimental grid with five random seeds per cell on BCI IV-2a. Logging all metrics, per epoch training curves, runtimes and per-trial inference latency. This produces 60 cell seed combinations that form the primary results. Repeat this on IV-2b and on 4 channel and 22 channel montages for IV-2a specifically for a wide array of results spanning 180 cell seed combinations.

11. **Statistical Analysis** Undergo statistical analysis, paired tests across seeds for architectural comparisons, pipeline comparisons. Analyse confidence intervals, effect sizes, multiple comparison correction using a documented method. The analysis must follow the plan registered prior to grid execution.

12. **Discussion, conclusion and revision** Discussion and conclusion should be heavily drafted nearing completion. Points should be sufficient and coherent with every claim tracible to relevant evidence and supported by reproducibility as well as figures and tables.

13. **Final submission** Every aspect of the project should be completed with no errors or omissions. A video demonstration is required 3 days after the initial submission.

1.4 Project risks

This section identifies all risks considered in the form of a risk register.

1.5 Time plan

The project runs from 1st June to 28th August 2026. It contains six phases with a critical path running through the data loader, evaluation framework, and EEGNet baseline. There is a time buffer for each task in order to account for unexpected issues. Six milestones mark the major checkpoints: The evaluation framework; all three architectures baseline achieved; completion of the experimental grid; completion of statistical analysis; the full draft of the dissertation and the final submission.

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Phase 1: Foundation & data												
1.1 Environment setup												
1.2 Dataset download												
1.3 Data loader												
Phase 2: Preprocessing & evaluation												
2.1 Preprocessing config												
2.2 Determinism tests												
2.3 Evaluation frame												
Phase 3: Model development												
3.1 EEGNet												
3.2 EEG-Conformer												
3.3 GRFNet												
3.4 Training interface												
Phase 4: Grid execution												
4.1 Grid execution												
Phase 5: Stats tests and figure generation												
5.1 Statistical analysis												
5.2 Figures & tables												
Phase 6: Report development												
6.1 Discussion & revision												
6. 2 Proofreading & submission												
Milestones			**M1**			**M2**		**M3**	**M4**		**M5**	**M6**

Figure 1.1: Final Gantt Chart

Table 1.1: Risk register: initial rating, mitigation and residual rating

ID	Risk	Initial (L/I)	Mitigating action	Residual (L/I)
R1	Failure to replicate model architecture	M/H	Start with EEGNet to validate the loader before the more complex models	L/H
R2	Irrelevant or inconclusive findings	M/L	Reproducible framework and galea results provide a valuable contribution	L/H
R3	Data loss or hardware failure	L/H	Push to Git after every session	L/L
R4	Misinterpreted statistics	M/H	Pre-register analysis plan with tests, thresholds and corrections	L/M
R5	Codebase faults	L/M	Work within a virtual environment with pinned requirements	L/L
R6	Computational resource restraints	L/M	Use local GPU as a primary source with a fallback to NTU computing	L/L

Chapter 2

Literature Review

Electroencephalography (EEG) is an invasive neuroimaging technique that is dominating brain-computer interface (BCI) research and cognitive neuroscience. Applying deep learning to EEG has been able to substantially develop signal analysis through the use of convolutional neural networks (CNNs), recurrent neural networks (RNNs) and transformer architectures which have replaced more primitive techniques such as manual feature extraction and windowed architectures. [Craik et al., 2019, Lyu, 2026]

It will consider the methodological issues found in recent studies and how these can be addressed to further the field. [Naderi and Jahanian-Najafabadi, 2025] found accuracies on a single OCD dataset ranging from 74.52% to 100%, however this is not due to architectural differences, it indicates inconsistent choosing of preprocessing methods.

[Kessler et al., 2025] demonstrated that specific preprocessing pipelines caused the model to learn the specific structural patterns of noise for that one particular pipeline rather than learning the underlying patterns. [Geirhos et al., 2020] refers to this as "shortcut learning". This means they have no transferable potential and are therefore less relevant for wide scale deployment. This problem gets considerably worse when considering the cross-subject variability of EEG, A model trained on one group will have diminishing accuracy values when extrapolated to an independent participant.

2.1 Electroencephalography (EEG) Data

Electroencephalography (EEG) is an invasive neuroimaging technique that records the electrical activity of large synchronous clusters of cortical neurons. These voltage measurements are captured through electrodes placed on the scalp, providing a temporal resolution that is typically in the millisecond range. This is considered superior to other modalities such as Magnetoencephalography (MEG) and functional Magnetic Resonance Imaging (fMRI), despite MEG having comparable millisecond resolution.[Craik et al., 2019] The relatively low cost has established EEG as a fundamental technology within cognitive neuroscience research and brain-computer interface development.

2.1.1 Signal Acquisition and Electrode Systems

EEG signal collection follows a standardised electrode distribution. The most commonly used system is the 10-20 international system, which defines fixed positions on the scalp relative to key anatomical landmarks. Higher density systems such as 10-10 follows the same principle, allowing for reproducible signal collection and consistent scalp coverage across the cortical regions. [Klem et al., 1999] The EEGMMIDB dataset follows the 10-10 system with 64 electrodes.

2.1.2 Frequency Bands

Raw EEG signals are typically decomposed into frequency bands that are associated with distinct functional roles. These bands are important because they provide the foundations in which most feature detection pipelines operate. The main frequency bands are: Delta (0.5-4 Hz) which is linked to deep sleep; Theta (4-8 Hz) which is related to working memory and cognition; Alpha (8-12 Hz) which is associated with relaxing and related to attention; Beta (12-30 Hz) which reflects active concentration and motor function and gamma (above 30 Hz) which is relevant to cognitive processing and feature binding.[Adhikari et al., 2025] These are relevant because the power shift across these bands is key to distinguishing between tasks and conditions. For motor imagery decoding

specifically, the mu rhythm (8-12 Hz) and beta band over the sensorimotor cortex are the key bands to consider, the modulation of their spectral power relative to the baseline during motory imagery tasks allows for filter banking on common spatial pattern methods in order to increase BCI performance.[Chin et al., 2009]

2.1.3 Preprocessing Techniques

The issue with EEG data is the signal-to-noise ratio. To handle this, preprocessing techniques can be applied to clean the data and make the cognitive activity signals less ambiguous.[Khademi et al., 2023] Many things such as muscle contractions and a beating heart can cloud cognitive activity signals which we want to detect. this can be negated by band-pass filtering applied most commonly between 0.5 and 45 Hz as well as a notch filter which can be applied to reduce the hum of power lines at around 50 - 60 Hz. [K and K, 2026]

Common Average Reference (CAR) can be applied to suppress global signal components that are consistent across the scalp electrode positions. This is done by subtracting the mean of all the electrodes from each individual electrode. At every point in time you can calculate average voltage and subtract it from each individual electrode. The core principle is that relevant signals are spatially localised sufficiently enough that, subtracting common components across all electrodes removes irrelevant noise. [Tsuchimoto et al., 2021] used simultaneous EEG-fMRI to show that CAR produced an optimal signal-to-noise ratio for extraction of sensorimotor rhythm activity which indicates it has potential applications to the EEGMMIDB dataset.

Independent Component Analysis (ICA) is another technique that separates aspects of the signal into independent components. Tools such as MATLAB and MNE-Python are the main tools implemented for this preprocessing step. Interference such as eye blinks can be identified through this analysis method and removed without damaging the data.[Delorme and Makeig, 2004] A recent study found that ICA and other techniques reduced classification accuracy, the reason for this was inferred to be because models learned from the structure of noise rather than the independent brain signals which means

preprocessing choices are not natural.[Kessler et al., 2025]

2.2 Introduction to Machine Learning and Deep Learning

Machine learning is a derivative field of artificial intelligence and a broad term for algorithms with the purpose of pattern recognition. Approaching EEG data with deep learning rather than 'hard-coded' algorithms and parameters to identify EEG fluctuations is becoming the primary approach to EEG analysis because brains signals are too complex for a scalable 'hard coding' approach and therefore need flexibility in order to be implemented in BCIs. [Craik et al., 2019]

2.2.1 The Classical Machine Learning Approach to EEG

The classic machine learning pipeline for EEG classification was to preprocess raw signals in order to remove noise, hand crafted features were extracted from the cleaned data, most commonly Power Spectral Density, Event Related Potentials or Common Spatial Patterns (CSP). They were then passed to a classifier such as a Support Vector Machine (SVM) which was optimal for EEG data due to it's non linearity. SVMs utilised the kernel trick to handle this because it was able to implicitly map the input features in order for a linear separator to then be applied. Filter Bank Common Spatial Patterns (FBCSP) extended CSP across multiple frequency bands, which, combined with an SVM classifier [Ang et al., 2012], became standard for motor imagery classification.

2.2.2 Deep Learning Standards

Deep learning replaced manual feature extraction with learnable processing; a deep network was able to learn a hierarchy of abstract classes directly from the data. Deep learning used unsupervised pre-training to provide a baseline for the model. Restricted Boltzmann machines [Hinton et al., 2006] and stacked de-noising auto-encoders[Vincent et al., 2010] gave substance to this baseline and every layer was trained before supervised fine-tuning[Wulsin et al., 20

was undertaken.

Deep belief networks were applied to anomaly detection and sleep staging by [Wulsin et al., 2011] and [Längkvist et al., 2012], respectively, with convolutional neural networks following soon after, driven by developments in GPU-based training. The first widely used end-to-end CNNs for EEG, developed by [Schirrmeister et al., 2017, Lawhern et al., 2018], replaced feature extraction with learnable convolutions. A temporal convolution would act as the band-pass filter, with a spatial convolution acting as a CSP method. A final fully connected layer would be trained using cross-entropy loss.

2.2.3 Convolutional Neural Networks

With the emergence of more advanced deep learning techniques, the kernel trick is less relevant today, as algorithms such as convolutional neural networks can learn their own nonlinear transformations through convolutional layers. This process of reshaping allows for easier and more efficient classification because they are able to learn from the provided data rather than relying on the hard-coded predefined functions that have been applied previously.[Pušica et al., 2025] A single convolutional layer applies learnable filters to the input layer. The following equation, which is actually used within neural networks, is known as cross-correlation due to not being rotated and can be written as:

$$y(t) = \sigma\left(\sum_{k=0}^{K-1} w(k)\, x(t+k) + b\right)$$

[Goodfellow et al., 2016]

In this equation, x is the input signal, w is the filter with a small set of weights, b is the bias term and σ is the non linear activation function. In terms of EEG models, the exponential linear unit is most commonly used in order to avoid the dying neuron problem that ReLU can face. The filter weights and bias are learned from the data during training, this means as the neural network builds up representations of the input data, it can discover features by itself.[Craik et al., 2019]

2.2.4 Recurrent Neural Networks and Long Short-Term Memory

Recurrent Neural Networks (RNNs) operate on sequences and capture temporal dependencies rather than spatial patterns, as do CNNs. This is relevant to EEG task classification since brain activity unfolds over time, and prior signals can affect future ones. Standard RNNs suffer from the vanishing gradient problem, which diminishes the ability to learn long-term dependencies. A long short-term memory (LSTM) network counteracts this, as it has a continuous cell state that persists information across multiple steps, controlled by four gating mechanisms. This input gate equation expresses the principle:

$$f_t = \sigma(W^f \cdot [h_{t-1}, x_t] + b^f)$$

[Li et al., 2020]

These are the forget, input and output gate, respectively. x_t is the input at time t, $h_t - 1$ is the previous hidden state, w is weight, b is bias and σ is the sigmoid function that condenses values to between 0 and 1. The forget gate decides what to discard from the previous cell state, the input gate decides what new information to add to the cell state and the output gate decides what to output forward to the next time-step. The combination of CNNs and LSTMs into one single architecture is gaining traction because it covers spatial and temporal aspects of the data which is beneficial for multimodal data such as EEG because patterns can be found across multiple dimensions. [Lyu, 2026]

2.2.5 Training, Optimisation and Regularisation

Loss Functions and Back Propagation

Training a neural network involves adjusting weights and biases to minimise a loss function. The most commonly used loss function is cross-entropy and is computed as follows:

$$\mathcal{L} = -\sum_c y_c \log(p_c)$$

[?]

Backpropagation is an algorithm that computes the loss gradient in reference to every parameter in the network. It applies the chain rule from the output to the input, each weight has a back-propagation computation which indicates the direction and magnitude of change needed to reduce loss. The key benefit of back-propagation is it's ability to calculate all gradients in one single backwards pass.[Rumelhart et al., 1986]

Optimisers

An optimiser takes in these gradients and updates the parameters accordingly. Adam develops this by implementing adaptive learning rates for each parameter based on the mean and variance of the gradient history. Adam is the most common optimiser because of its robustness across a wide range of architectures.[Ruder, 2017, Kingma and Ba, 2017]

Over-fitting and Regularisation

Over-fitting occurs when a model learns patterns specific to the training data rather than the underlying patterns that can be generalised. This is a bigger overall problem in EEG research as datasets are commonly small which leaves a gap between model capacity and data quantity.

Dropout is the most common form of regularisation in deep learning. During a forward pass in training, a random fraction of neuron activations are set to 0, this prevents the network from relying on a single neuron. At inference time, all neurons are active however their outputs are scaled according to the dropout probability to maintain consistent activation magnitude. Dropout allows each training step to focus on different sub-networks, the full network at inference time gives an average prediction of an exponential number of sub-networks. [Srivastava et al., 2014]

Batch normalisation normalises activations within a mini-batch to have zero mean and unit variance at each layer, it then applies learnable scale and shift parameters, this stabilises training which allows for higher learning rates, it provides regularisation as there is minimal noise from the mini-batch.[Ioffe and Szegedy, 2015] A common problem

of batch normalisation is that the batch size usually has a large influence on it's effect, so with EEG's small dataset problem, batch statistics can become noisy and unreliable. Layer normalisation is able to combat this by normalising across the feature dimension rather than the batch dimension which therefore makes it preferable for EEG focused models. [Ba et al., 2016]

Self-Attention and Transformer Architectures

Transformer architectures are also increasingly relevant as of recent. They use a concept known as self-attention that allows a network to analyse all of the input data at once rather than step by step. This works by computing a weighted sum of all other positions, at each position, in which the weight reflects how relevant each position is to the current one. This process allows transformers to pick up more vague patterns in the data and can compute more efficiently than RNNs because of it. 'Spectro-temporal' transformers are already outperforming CNNs on tasks such as speech decoding and emotional recognition. [Milyani and Attar, 2025, **?**]

Convolutional Vision Transformers (CvT) introduced by [Wu et al., 2021] adds convolutions into the vision transformer, firstly to replace patch embedding with a convolutional token embedding, then with a convolutional projection inside each transformer block to replace linear projections. The key benefit of this is to combine the shift, scale and distortion invariance of CNNs with the dynamic attention and global context of transformers. Other convolutional transformers such as CTNet developed by [Zhao et al., 2024] achieved 82.95% on BNI IV-2a and 88.49% on BCI IV-2b in subject specific evaluation. EEG-VTTCNet developed by [Shi et al., 2024] reports 84.58% and 90.94% on the same dataset, utilising a vision transformer, temporal convolutional network hybrid model. MSCFormer developed by [Zhang et al., 2025b] achieved 82.95% and 88% with a convolutional front end and TCFormer ([Altaheri et al., 2025]) achieved a 84.79% on BCI IV-2a as well as 96.27% on the High Gamma dataset using grouped-query attention.

As of recently, [Song et al., 2023] combined a convolutional patch embedding with a transformer encoder for motor imagery decoding, achieving 78.66% on BCI IV-2a and 95.30%

on SEED. [Wang et al., 2025] develops this by adding a dual-branch design that processes both spatial and temporal information together, massively reducing the parameter count while also improving accuracy.

CvT architectures are currently considered some of the best performing models for motor imagery, however there have been developments that push this standard further.

Selective State Models

Substituting self attention with selective state space models such as Mamba is a key aspect of some recent studies. [Gu and Dao, 2024] This achieves linear complexity in sequence length which is a benefit to longer EEG recordings. MI-Mamba[Guo et al., 2025] combines a single convolutional layer for spatial feature extraction with a Mamba module for global temporal modelling, achieving 80.59% on a four class BCI IV 2a dataset and 84.42% on BCI IV-2b. while reducing the parameter count considerably. EEG-ConvMamba[Huang et al., 2026] develops a multi branch CNN that suppresses noise and reducing sequence redundancy before passing a compact feature sequence to the Mamba module.

ROCKET/MiniRocket

Random convolutional kernel transform generates a bank of random kernels with fixed weights, it applies them to time series input and extracts two statistics (PPV, maximum activation) and then feeds the vector it outputs to a ridge regression classifier [Dempster et al., 2019].

The kernels, however, are not updated via gradient descent, only parameters in the linear head are learned. This allows for much more efficient training while still matching or exceeding the accuracy on time series benchmarks.

[Dempster et al., 2021] proposed MiniRocket, this replaces the random sampling with small deterministic sets of kernel configurations (fixed length, bias terms and dilations) which, once again, cuts training times and variance. The key architectural distinction is that feature extraction is fixed so it can't overfit which is extremely optimal for smaller

datasets which are common in EEG task classification.

2.3 Applications of Machine Learning to EEG Data

The main areas of machine learning application to EEG include: motor imagery classification; Emotional recognition; the diagnosis of diseases as Alzheimer's and Parkinson's as well as conditions such as OCD and depression; mental workload estimation, seizure detection and sleep staging.[Craik et al., 2019]

2.3.1 Motor Imagery Classification

Motor imagery is a key aspect of BCI research. Early approaches relied on event-related features that targeted de-synchronisation in the mu and beta bands, these were then classified using SVMs. As of recently, [Titkanlou et al., 2025] tested CNN-LSTM and CNN-Transformer models on a motor imagery dataset of 29 subjects, they received a 79% accuracy result with the CNN-LSTM. They also tried data augmentation using noise injection, conditional variational auto-encoders and Wasserstein Gans to deal with the main issue of EEG datasets which is that they are too small and therefore have low external validity.

Motor imagery evaluation uses one method, subject specific which trains the model and tests the model on one subject. Many papers opt to use subject specific evaluation which usually results in an improved accuracy of around 10-20% compared to cross subject which leaves room for potential challenges to be addressed for large scale BCI implementation. The primary dataset for motor imagery classification is the BCI IV-2a Competition dataset, this revolves around nine subjects performing four key motor imagery tasks and uses 22 EEG channels sampled at 250 Hz with two sessions of 288 trials. The BCI IV-2b dataset includes three bipolar channels for easier sanity checking. [Tangermann et al., 2012]

2.3.2 Other applications

Emotion Recognition

EEG Emotion recognition is driven by datasets such as DEAP and SEED. Studies commonly use spectral features from the theta, alpha and gamma bands with spatial attention in order to classify emotional states. CNN-LSTM models tend to be $\tilde{8}0\%$ accurate on SEED for multi-class problems, ablation studies show that using CNNs or LSTMs alone does not match CNN-LSTM models.[Zhu et al., 2024] Transformers methods, however, have achieved an accuracy of 96.1% [Lyu, 2026]

Cross dataset generalisation remains a prominent issue, especially for emotion recognition. Models trained on SEED perform significantly worse when applied to DEAP, the same applies both ways. This is due partly to differences in test stimuli, but also suggests domain specific accuracy values which do not reflect a generalised representation of the target classification, but rather the models capacity to learn the training data.

Alzheimer's, Parkinson's and OCD

For Alzheimer's detection, CNN-LSTM hybrids have been used for early detection by analysing spatial patterns within slowed brain rhythms and a decline of signal complexity. [Lyu, 2026] A study on Parkinson's classification found XGBoost, which is an open source ML library that implements optimized distributed gradient boosting, outperforms deep learning models on a small dataset, achieving an accuracy of 79%. [Avola et al., 2025] For OCD, [Naderi and Jahanian-Najafabadi, 2025] found that studies using the same dataset ranged from 74.52% to 100%, the difference was identified to be the design of the model, preprocessing and validation rather than any relevant methodology advances.

A common methodological criticism is the small group sizes within clinical papers, with regards to machine learning it results in wide confidence intervals which indicate uncertainty and low precision. Depression and Anxiety classification face the same issues which results developments based in dependence on larger groups rather than architectural development. [Avola et al., 2025] reporting on Parkinson's found that a traditional gradient boosted classifier outperforms the current deep models.

Mental Workload Classification

Mental workload classification is relevant for neuro-ergonomic applications such as monitoring pilots. The common findings is that frontal theta power increases and parietal alpha power decreases as cognitive demand increases. [Pergher et al., 2019] The issue that commonly presents is the spectral features are not accurate enough to distinguish between intermediate workload levels. Machine Learning classifiers are able to handle non-linear patterns and therefore can be used to assist the classification. [Bjegojević et al., 2024] [Pušica et al., 2025] found that it was difficult to compare results as there was large variation across studies in terms of preprocessing, features, segment lengths and validation. The inconsistency in methodology represents a larger reproducibility problem within the field, preprocessing choices can have large variations from study to study with reported accuracy being unreliable for evaluating a models capabilities. As well as this, mental workload classification is very deterministic in the sense that, workload specifically is complex and continuous, so when the classification consists of only 3 classes (low, medium and high) it may be missing more nuanced aspects. Deep learning approaches are applied to capture non linear spatial and temporal structures of workload related activity, however the field is still less developed compared to something such as motor imagery due to there being fewer 'benchmark' datasets [Pušica et al., 2025]

2.4 Metrics for Evaluating Machine Learning Performance

2.4.1 Key metrics

Accuracy

Accuracy is the most common metric and is measured by the percentage of all predictions that are correct. It is commonly computed as follows:

$$\text{Accuracy} = \frac{tp + tn}{tp + tn + fp + fn} \tag{2.1}$$

The problem with accuracy is that it can be misleading if classes are imbalanced. For motor imagery datasets, rest periods account for a disproportionate amount of the recording time. This means that a model that predicts rest for large amounts of epochs can achieve a high accuracy without learning any actual motor imagery features.

F1 metric

The F1 score combines precision and recall into a single value using a calculation known as the harmonic mean and is computed as:

$$\mathrm{F1} = 2 \times \frac{Precision \times Recall}{(Precision + Recall)} \tag{2.2}$$

The harmonic mean is used to replace a standard average as it roots out anomalous results. A classifier that detects T2 with a precision of 95% but only a recall of 10% the mean would be 52.5%, [Varoquaux and Colliot, 2023] however the harmonic mean would be 18% which better reflects the dimensions of the model. A limitation of F1 is that it ignores true negatives (correctly identified rest epochs) which means it misses out on key aspects if rest occupies a large portion of a dataset (which it generally does).

Cohen's *k*

Cohen's k measures whether predicted and true labels are correct for chance. [Cohen, 1960]

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2.3}$$

Where an observed and expected accuracy by chance is given, a classifier predicts that predicts only the majority class achieves $k = 0$ regardless of the accuracy because the prediction matches the chance baseline. This is key for identifying a class collapse and is a key reason for using it as a complementary metric.

Matthews Correlation Coefficient

The Matthews correlation coefficient (MCC) [Chicco et al., 2021] is an emerging measure as it takes into account all four values from the confusion matrix and is computed as follows:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{2.4}$$

MCC returns a value between -1 and 1, where 1 is perfect, 0 is average, and -1 means every prediction is wrong. Unlike the F1 metric, MCC accounts for all four matrices, providing a complete performance measure when a class, such as rest, accounts for a large proportion of the data. [Varoquaux and Colliot, 2023]

2.4.2 Other metrics

Precision and Recall

Precision and Recall metrics approach this differently. Precision addresses how many epochs the model flagged as a certain motor imagery class were actually of that class. For example, how many epochs that were classifying imagined left-hand movement were actually left-hand imagery.

$$\text{Precision} = \frac{tp}{(tp + fp)} \tag{2.5}$$

A high FP, which would mean the model is mislabelling rest epochs as active motor imagery, ultimately decreases precision.

$$\text{Recall} = \frac{tp}{(tp + fn)} \tag{2.6}$$

A high FN count decreases recall, indicating that the system fails to identify actual motor imagery. This is why a balance must be identified. A model flagging epochs as positive will increase recall but decrease precision.

ROC curve analysis

A ROC curve plots the true positive recall rate against the false positive rate across all possible decision thresholds:

$$\begin{aligned} \text{True Positive Rate} &= \frac{tp}{(tp + fn)} \\ \text{False Positive Rate} &= \frac{fp}{(fp + tn)} \end{aligned} \tag{2.7}$$

The area under this curve (AUC-ROC) gives a single value for how well the model separates classes. An AUC of 1.0 means the model distinguished perfectly, whereas 0.5 means it is at chance performance.

Confusion Matrix

Four values are derived from this matrix; true positive (TP), true negative (TN), false positive (FP) and false negative (FN).[Varoquaux and Colliot, 2023] In the context of MI classification, TP would be when the model correctly identifies a movement or imagery epoch. A TN is when the model correctly identifies rest or a baseline epoch with no active motor function. An FP is when the model incorrectly identifies a rest epoch as an active task, this would be identifying T1 during a rest period and a FN is when the model fails to detect motor imagery epochs, resulting in it being labelled as rest.

2.5 The OpenBCI Galea Headset

OpenBCI's Galea is a consumer grade multimodal biosensing headset. The key physiological modality, EEG, provides 10 channels (F1, F2, C3, Cz, C4, P3, Pz, P4, O1 and O2) recorded through dry soft polymer active electrodes, which do not require any gel and are designed to be equipped simply and efficiently outside of controlled research settings.
The electrode positions follow the 10-20 system but coordinate positions do tend to vary and are relatively hard to document reliably. This spans from the frontal, central, parietal and occipital regions including the sensorimotor strip (C3, C4, Cz) which is key for motor imagery decoding. However dry polymer electrodes lack hair penetrating geometry which

results in a higher impedance across hair covered areas as well as an elevated line noise.

2.6 Open Challenges

2.6.1 Preprocessing-Architecture generalisation

One of the biggest challenges for EEG motor imagery classification is the preprocessing pipelines. Many choices can affect reported performance across a variety of models: picking which preprocessing techniques to use; which techniques positively or negatively affect the model, and so on. [Kessler et al., 2025] reported that decoding accuracy is affected by preprocessing choices. As mentioned before, where the structure of the specific pipeline is learnt rather than the underlying patterns.[Naderi and Jahanian-Najafabadi, 2025] observed impossible accuracies of 100% on one single dataset and attributed it to inconsistent preprocessing choices rather than sufficient architectural development. Papers that introduce new architectures rarely test them across multiple preprocessing pipelines which make it difficult to determine whether the reports indicate actual architectural advantages or are simply favourable preprocessing of this specific model.

2.6.2 Cross-Subject generalisation

A common problem with EEG signals is that when a model is trained on a specific set of subjects, it struggles to generalise its learning to extraneous individuals. This is a large barrier for wide scale BCI's that aim to be un-calibrated and yet still functioning and deployable for real world applications. [Habashi et al., 2025] The most common evaluation method is subject-specific, reported accuracies in the subject-specific method is consistently higher with regards to accuracy. [Zancanaro et al., 2021] reported that this gap is large enough that an architectures supposed advantage over a baseline model is only identified once cross-subject validation is undertaken. Techniques such as transfer learning and domain adaption are presented to reduce this gap[Zheng et al., 2025] but are rarely evaluated consistently which can make reported findings interpretable and unreliable.

2.6.3 Cross-Dataset validation

Cross dataset generalisation is one of the largest barriers in EEG. Datasets differ in their data collection with regards to hardware, channel montages, sampling rates and various other factors, this can mean the difference of performing well on one dataset and poorly on another. Most published architectures aren't tested on vastly different datasets. [Zhang et al., 2025a] demonstrates the difficulty of transferring from a 62 electrode dataset and an 8 electrode one. Some authors explicitly state that they ommit cross-dataset validation in order to preserve comparability with prior single-dataset studies. [Hwaidi and Ghanem, 2026a]

2.6.4 Proposed Direction

These challenges are rooted in uncontrolled preprocessing, evaluation and reporting. It is therefore proposed that this paper is to develop a controlled benchmark study focussed on analysing the statistical and reported differences through different preprocessing and architectural choices. The study will pair a three architectures and four preprocessing pipelines with one evaluation methods into a controlled experimental grid. This can reveal whether these architectures have tangible advantages across multiple pipelines or if the model favours the shape of the signals it was trained on. Utilising these protocols alongside releasing the full codebase and configuration specifically to focus on creating a reproducible benchmark, this study will directly address some of the largest concerns in EEG.

Chapter 3

Methodology

3.1 Aim and design principles

There are three key principles governed by the design decisions. Firstly, *reproducibility*: Everything that outputs a result is recorded per-run via a manifest and folded into a configuration fingerprint. Second, *honest result reporting*: The hardware constraints remove information which is likely to reduce the evaluation metric values, this loss is reported as a result rather than compensated for with hidden reconstruction. Third, *faithful hardware implementation*: Every stage of the pipeline should only see the signals recorded by the electrodes that match the Galea headsets in order to report results that are relevant to Galea rather than higher level research headsets.

3.2 Requirements and design strategy

These requirements roughly follow from the open challenges (section 2.6): the preprocessing pipelines and common model interface respond to the preprocessing architecture confound; the subject specific protocol responds to the cross subject generalisation gap; and the manifest and fingerprinting responds to the reproducibility.

Type	Requirement	Design response
F	Load BCI IV-2a/2b (GDF) and restrict to the Galea montage	Name-normalised loaders and channel selection in the **data** layer
F	Four preprocessing pipelines of graded modulation	A YAML pipeline of composable steps
F	Three interchangeable decoders under one interface	A model interface and a registry
F	Subject-specific evaluation on four metrics	A leakage-safe split and shared metrics
F	Complete provenance for every run	A per-run manifest and an order-independent fingerprint
NF	Reproducibility and determinism	Single-seed control, a determinism test and a pinned environment
NF	Leakage-safety	Splits verified disjoint by construction and re-checkable
NF	Extensibility to new models and pipelines	Additions require no change to the shared code
NF	Portability across montages and sampling rates	Model dimensions derived from channel count and rate

Table 3.1: F = functional, NF = non-functional

3.3 Development methodology

The framework was built under a test-first principle with phase gating: functionality is only accepted when its tests pass, so each gate: data and preprocessing, leakage safe evaluation, the baseline and shared trainer, the architectures and the grid, all have explicit success criterion. No metrics are reported before the splits, units and provenance are verified.

Testing Hermetic unit tests check pure logic, dependency gated integration tests determine the functionality of MNE, PyTorch and dataset paths and skip cleanly when they aren't relevant, and end to end tests to reproduce published baselines via scripts, are all included as key testing operations. Each levels enforces a system contract, each model is analysed on: the forward and backwards pass, the output shape, the parameter count, and the rejection of erroneous inputs; the trainer is analysed on determinism from seeding and atomic output and the pipeline is analysed on its bit-identical preprocessing between runs.

3.4 Resources required

1. **Tools and Technologies** Python 3.11 or later is optimal, the leap in efficiency between 3.10 and 3.11 warrants this minimal version. All three models are implemented in PyTorch. EEGNet, published in Keras/ TensorFlow, and EEG-Conformer were reimplemented here as PyTorch ports as it allows for more control over architectural decisions. MNE-Python will be used as it allows for EEG-specific preprocessing. NumPy and SciPy for calculations, Scikit-learn for metric analysis, Configurations are declared in YAML and dependency versions are pinned for reproducibility, PyTest for unit testing and finally git for version control. For signal acquisition, the Galea GUI provides sufficient signal analysis tools

   With regards to storage, optimally 50-100GB of storage to be available on any workstation for the whole project, including datasets, models, logs, snapshots as well as to be cautious for memory leaks and experimentation.

2. **Hardware** Access to Open BCI's Galea Headset will be required for raw data capturing, this is provided by NTU and will be the primary method of data collection.

   A computer with a CUDA compatible GPU will be required. For deep learning standards, a 30-series or above GPU with compute capabilities upwards of 8.0 with 8-16GB of VRAM is optimal. The personal computer used exceeds this standard and will be my primary workstation, with a secondary workstation within NTU's computer lab and a tertiary workstation being a CUDA GPU cloud instance such as Google Colab.

3. **Datasets** Two publicly available motor imagery datasets from BCI Competition IV are used. [Brunner et al., 2008, Leeb et al., 2008] Dataset 2a contains nine subjects, 22 EEG channels (3 EOG), four classes (left hand, right hand, feet, tongue), sampled at 250 Hz. Each subject relates to a training session (T) with cue labels in the recording and evaluation session (E). Dataset 2b contains nine subjects, 3 EEG channels, two classes (left and right hand) sampled at 250 Hz, with five sessions per subject (3 (t), 2 (E)).

4. **Supervision** Meetings every 2 weeks, as well as on specific milestones laid out in the time plan.

Dataset file structure and epoching Recordings are read from GDF files with MNE-Python[Gramfort et al., 2013] and the EOG channels are dropped. For evaluation sessions the withheld class labels are loaded from label files and aligned to trials by cue label. Continuous recordings are epoched around each cue using a fixed window of $\text{round}((t_{\max} - t_{\min}) \times f_s)$ samples, where f_s is the sampling rate. The default window is 0-4s post cue, the window is a configurable recorded parameter.

3.5 Electrode montages constraints

The main methodological constraint is that every dataset and model limited to the ten Galea EEG channels: F1, F2, C3, Cz, C4, P3, Pz, P4, O1 and O2. The datasets used only contain part of this electrode set and therefore will be used as an alternative to a padded superset.

Table 3.2: Galea montage restriction: each dataset only contributes its overlap with the ten galea electrodes

Dataset	Native channels	Retained (Galea overlap)
IV-2a	22	4: C3, Cz, C4, Pz
IV-2b	3	3: C3, Cz, C4
Galea recording	10	4: C3, Cz, C4, Pz

This overlap is used because in order to fabricate absent electrodes, it would require injecting interpolated signal which would misrepresent the capabilities of the headset. Channels are selected by name and returned in canonical order so that the channel axis is deterministic across datasets. Selection is applied before preprocessing so the montage dependent operations are computed only from the relevant electrodes.

The galea restrictions create a distinct variant of this public benchmark, where the reduction from 22 channels to 4 with remove coverage that spatial filters tend to exploit which therefore results in a drop in accuracy. This itself is a reported finding and will be discussed. The original architectures will be reproduced with the full 22 channel montage

in order to achieve the reported results so that the parameters are the same and are reproducible when constrained by the galea overlap.

3.6 Preprocessing modulation gradient

Four preprocessing pipelines have been defined for the purpose of analysing the effect of preprocessing strength on decoding performance, each pipeline operates on a continuous signal before epoching.

Table 3.3: The four preprocessing pipelines, forming a graded modulation axis.

Level	Operations	Role
minimal	Band-pass 0.5–100 Hz	Near-raw floor; retains drift-free broadband signal, no amplitude normalisation.
standard	Band-pass 4–40 Hz, per-channel z-score	Reproduces the preprocessing of the reference architectures.
deep	Band-pass 8–30 Hz, average reference, resample to 125 Hz, z-score	Tight motor band, spatial referencing, temporal decimation.
maximum	*deep* plus a Hjorth surface Laplacian	Heaviest spatial shaping (one montage-adaptive operator).

3.6.1 Amplitude normalisation and spatial transformation

Per channel z-scoring is applied within the pipeline using each recordings statistics. This is unsupervised and per recording in order to be leakage safe for train/evaluation.

Hjorth laplacian

The *maximum* pipeline applies a spatial sharpening operator to every montage, [Hjorth, 1975] introduced a surface Laplacian transformation. This approximates the surface Laplacian, which estimates the second spatial derivative of the scalp potential through a linear operator, each electrode is re-referenced against an inverse-distance-weightred average of it's nearest neighbours. The neighbours follow directly from the distances between electrode coordinates and their weights sum to one so that the transform cancels any spatially uniform components and retains only the varying activity. This allows the operation to

occur regardless of channel count which was a previous constraint and has been discussed in 3.7.

3.7 Determinism, and why not ICA?

The choice for maximal spatial transformation was constrained by the requirement of reproducibility. Independent component analysis (ICA) was rejected because of this, it's decomposition is initialised from a random state, so the recovered components and any signal that is reconstructed after removing them will vary between runs unless the seed is fixed and written to a manifest. Despite this proposed solution, identical output is still not guaranteed across hardware. Threading a stochastic dependency through the preprocessing stage would undermine the principles this study aims to provide.
The *maximum* modulation stated in 3.3 are all closed form operations that produce bit identical output, this combined with the fact that not only does the Laplacian not require seeding, it is much cheaper to compute and is more compatible with the Galea montages than ICA as it is well defined on sparse montages based on montage adaptive neighbour graphs.

3.7.1 Pipeline gradient validation

The pipelines were verified to be distinct based on increasing gradients measured as the fraction of spectral power in the mu and beta bands and the mean absolute inter-channel correlation. Each level shifts these values in the expected direction which confirms that the levels of modulation intensity are genuine and sufficiently varying.

3.8 Model architectures

3.8.1 EEGNet

[Lawhern et al., 2018] published the first of three decoders, EEGNet. it is reimplemented via PyTorch following the reference architecture. A temporal convolution which acts as

a set of learned band pass filters, a depth-wise spatial convolution that learns electrode weighting for each temporal filter and a separable convolution, all followed by batch normalisation. An ELU activation, average pooling along time and dropout, max norm weight constraints on the depth-wise and classifier weights and a final linear classification layer. The number of input channels it taken from the data so the architecture evolves around the montage selection in the configuration without having to alter any code. With respect to the reference guidance, the temporal kernel length is set to half the sampling rate and is therefore derived from each pipelines output rate rather than a hard-coded value.

3.8.2 EEG-Conformer

[Song et al., 2023] developed and published EEG-Conformer, for this project it was ported to PyTorch and checked against brain-decode to verify authentic porting. It involves a convolutional patch embedding stem which turns each trial into a token sequence which is then passed to a six layer transformer encoder with ten head self attention, it is then flattened and classified via a multi-layer perceptron. Porting was verified, on the 22-channel 1000-sample input reference the stem yields 61 tokens and the model has 789, 572 parameters. There are three documented deviations where necessary: The spatial convolutional kernel and head width are derived from the montage rather than being hard-coded like the reference so that identical models run on the 3 specified deviating montages; the attention scaling reproduces the references full embedding factor; and the unused classification token branch and its fault optional mask path are replaced by a correct key mask.

3.8.3 GRFNet

The current novel contribution is **GRFNet**, it is a compact motor imagery decoder tailored specifically to the sparse Galea montage. It's key features regard low overfitting. [Dempster et al., 2019, Dempster et al., 2021] developed the ROCKET/MiniRocket concept where a largely fixed bank of random convolutional kernels feeds a linear head. The

models are known for the strong time series accuracy and low overfitting which is key for sparse EEG data, the published and library implementation do not include spatial filtering, pools only the proportion of positive values statistic, and uses a ridge classifier. GRFNet applies ROCKET's low overfitting recipe and makes it EEG, montage and task aware.

Novelty

GRFNet is inspired by ROCKET/MiniRocket. To distinguish this from being a re-implementation there are three key additions that tailor it to both EEG data and the Galea headset: a geometry aware learnable spatial front end, which supplies the spatial filtering that is key to motor imagery; a multi statistic fusion pool that combines PPV (ROCKET's temporal feature) with log variance band power (the FBCSP mu/beta event related synchronisation feature ([Schirrmeister et al., 2017])) and the mean; and an end to end trainable compact fusion head with per stream gating instead of ROCKET which uses a fixed ridge classifier. This model as a compact, low overfitting decoder tailored to the Galea headset is the key novelty.

Architecture

The pipeline maps an input trial of (batch, $n$channels, $n$times) to class logits through four stages. Firstly, *geometry aware spatial front end* which provides a learnable 1×1 convolution over the montage which projects the C channels to 16 spatial components, followed by batch normalisation and smooth spatial geometry seeding build from a 2D projection of the electrode coordinates. Secondly, a *ROCKET* style feature band with multi dilation depthwise 1D convolutions, it has fixed mean centred random kernels to preserve the key feature of low overfitting. Thirdly, *fusion pooling* where each feature map is summarised over time by three concatenated statistics. A differential PPV, log variance band power and the mean which derives from the multi pooling idea proposed by [Tan et al., 2022]. Finally, a gated compact head which regards a learnable softmax gate over the concatenated statistics, dropout and a linear classifier.

Parameter count and budgets

In the 22 channel configuration, GRFNet has 3655 trainable parameters, ROCKET kernels are excluded from the trainable count. This allows the model to be extremely compact, comparable to EEGNet's 3932 and significantly smaller than EEG-Conformer's 789,572. This allows for low overfitting which is a key aspect of this model and its objectives. Taken from [Schirrmeister et al., 2017], batch normalisation is applied to the feature maps before pooling rather than applied to the pooling vector. This preserves per trial band power discriminability.

3.9 Evaluation Protocol, splits and metrics

In the *subject specific* protocol, each subjects model is trained on that subject's training session and tested on their evaluation session.

3.9.1 Splits

Splits are constructed to be leakage safe: train, validation and test partitions are verified disjoint at the trial level. Each grid cell is repeated over five seeds to identify variability. aggregate means are stable across seeds while per subject scores are seed sensitive so that per subject results are reported as a mean and standard deviation over seeds rather than a single run, and architecture differences which are smaller than the seed spread are tested with paired statistics across seeds.

3.9.2 Metrics

The four metrics are chosen to give variety other than accuracy, they are collected as a group because on their own, each is ineffective at describing the honest results and their meaning. A defective model can score 0.25 on accuracy which would hide a class collapse, this is why Cohen's k and MCC are reported as they would identify it with a score of 0.0. Macro F1 displays a trend score of 0.1 on class collapse but this is not definitive evidence the way the others are.

3.10 Reproducibility and provenance

The study is designed to be reproducible, all stochastic sources namely, the Python, NumPy, PyTorch functions and the training data loader are seeded from a single per run seed with cuDNN in deterministic mode and no stochastic preprocessing is used, so the preprocessing stage is bit reproducible and a determinism test asserts identical output across reruns; CPU runs are bit identical, GPU runs remain reproducible only up to floating point non-associativity, since fully deterministic CUDA kernels are not enforced. Every reported figure is traceable to the exact configuration, data split and montage all traceable to a per run manifest and configuration fingerprint.

3.11 Validity issues and initial limitations

There are 22 channels in the IV-2a montage, and 10 channels on the Galea headset as shown in 3.2 which leaves an overlap of 4 channels, this displays limitations in the achievable spatial processing and reduces the separation between the higher preprocessing levels. The average reference of *deep* and the Hjorth surface Laplacian of *maximum* have only four channels to additionally modulate so the gradient is minimal unless used on the full 22 channel montage. This limit is caused by the number of electrodes: the same Hjorth operator is applied to every montage for consistency, with only four channels, each electrode has just three neighbours so the Laplacian sharpens the signa less than on a more dense montage. The decoders learn their own spatial filter and face the same shortage of channels, the accuracy ceiling is therefore set by the value of the available electrodes rather than the processed signal.

3.11.1 Artifact handling

Artifact removal (ICA) and bad channel repair are not applied to the datasets for reasons which are fundamental to the design and validity of results. Firstly, ICA will struggle on four channels from the Galea overlap because no meaningful source separation is possible. Secondly, [Bigdely-Shamlo et al., 2015] found that only non galea channels were

ever flagged, the four Galea channels (C3, C4, Cz, Pz) were all clean across all sessions when flaggin channels by robust log-variance deviation and cross-channel correlation. This points to the accuracy gap therefore being attributable to the reduction in spatial coverage not to the *bad* electrode noise.

3.12 The benchmark grid

The design ensures that every architecture is run under every preprocessing pipeline on both the four galea overlap channel and also the original 22 channel montages.
For example, GRFNet will pass through the four preprocessing intensities across five seeds. Based on the results of this, inferences can be made as to the functionality and relevance of certain modulations and their importance, or lack thereof. Three models by four preprocessing pipelines by five seeds creates 60 individually trained model for one montage on one dataset.

3.13 Statistical testing protocol

The raw seed accuracies are not sufficient for statistical comparison, five seeds per cell exist to stabilise the accuracy, collapsing to per subject means yield a single accuracy for each subject under each experimental condition.
Because the BCI IV-2a dataset contains nine subjects, each pairwise comparison relies on their matched observations which will dictate the choice of test.

3.13.1 Distribution-free tests

Given the small subject density, normality is difficult to assess, as well as this, classification accuracy is bounded to [0,1] and its sampling distribution is left skewed at high accuracy which violates the symmetry assumed by the paired t-test. BCI illiteracy is where almost a third of users produce near chance signals, this inflates inter-subject variance, distribution free rank tests make no distributional assumption and are therefore a sufficient choice based on the parameters identified here.

3.13.2 Friedman rank test

A Friedman test is applied when three or more matched conditions are to be compared. Each of the nine subjects is ranked across the conditions, the test statistic evaluates whether mean ranks differ more than chance predicts.
Pairwise follow ups are performed at certain threshold to avoid spending statistical analysis time on unwarranted comparisons.

3.13.3 Wilcoxon signed-rank tests with Holm-Bonferroni correction

Each pairwise follow up uses a Wilcoxon signed rank test which ranks absolute differences between matched pairs and compares positive against negative rank sums. Pairs per family (p-values) are corrected by the Holm-Bonferroni step down procedure which orders the raw p-values and compares each against a threshold.

3.13.4 Confidence intervals

Each mean accuracy difference is supplemented by a 95% bias corrected and accelerated (BCa) bootstrap confidence interval with 10,000 resamples. The BCa method corrects bias and skewness in the distribution.

3.14 Primary data collection protocol

Three Galea recordings were collected on the 25/08/2026 from three participants (A, D, J). The recordings aimed to recreate the BCI Competition IV-2a paradigm so that the recordings can be considered relevant and valid data for models trained on IV-2a and IV-2b. The Galea headset was sampled at 250 Hz, matching both dataset sources.
There are two deviations from the IV-2a protocol. Firstly, cue order is repeated in order in all three recordings rather than randomised, while this does leave the issue of anticipation, the further discussion will ultimately render these extraneous variables nullified. Second,

Recording	Eyes open	Eyes closed	Eye movement	Trials/class	Total trials
A	120 s	58 s	64 s	10	40
D	128 s	62 s	62 s	7	28
J	64 s	75 s	68 s	7	28

Table 3.4: Galea recording summary - Calibration blocks and motor imagery trial counts per participant

participant J's eyes open block is 64s rather than 120s, and J's cue pacing is irregular, deviating 4 seconds across all cues, whereas A and D deviate 0.83s and 0.46s respectively. The galea recordings retain EOG, facial EMG and forehead channels meaning the calibration block supports regressive ocular correction with respect to IV-2a's description. However this was not applied as the training grid presented in this dissertation drops EOG channels so implementing the Galea data would hinder the overall results due to the domain shift. Any implementation of symmetric corrections would result in the entire grid needing to be retrained which is outside the scope of this project.

Each session opens with the calibration block of: two minutes eyes open (marker 2), one minute eyes closed (marker 3), one minute eye movement (marker 4). The motor imagery segment delivers cues for tongue (5), left hand (6), right hand (7), and both feet (8). Trials are sliced into 4 second windows following each cue.

Chapter 4

Implementation

This chapter explains how the framework was built as a layered Python package (*eegchtb*), and how its engineering supports reproducibility which is a key aspect of the methodology. The sections below document the whole system, components, practicality and testing.

4.1 Package overview

Each concern in *eegchtb* has a single owner, it has a dependency light data layer, a preprocessing layer, an evaluation layer, a models layer, a training layer and an experiment module which all assemble into eegchtb-run. Every result is produced by the same path - Load, Preprocess, epoch, split, train, predict, score, record.

The layered structure follows the separation of concerns principle in which each sub package has a single responsibility and depends only on those beneath it, coupling is purposely kept low and every layer is independently unit testable.

4.2 Design decisions

Implementation choices are key in the development. First, EEGNet and EEG-Conformer are re-implemented as PyTorch ports behind an interface and result schema, meaning that all models share identical loading, splitting, metrics and manifests and differ only where strictly necessary. Second, every model derives its spatial and temporal dimensions

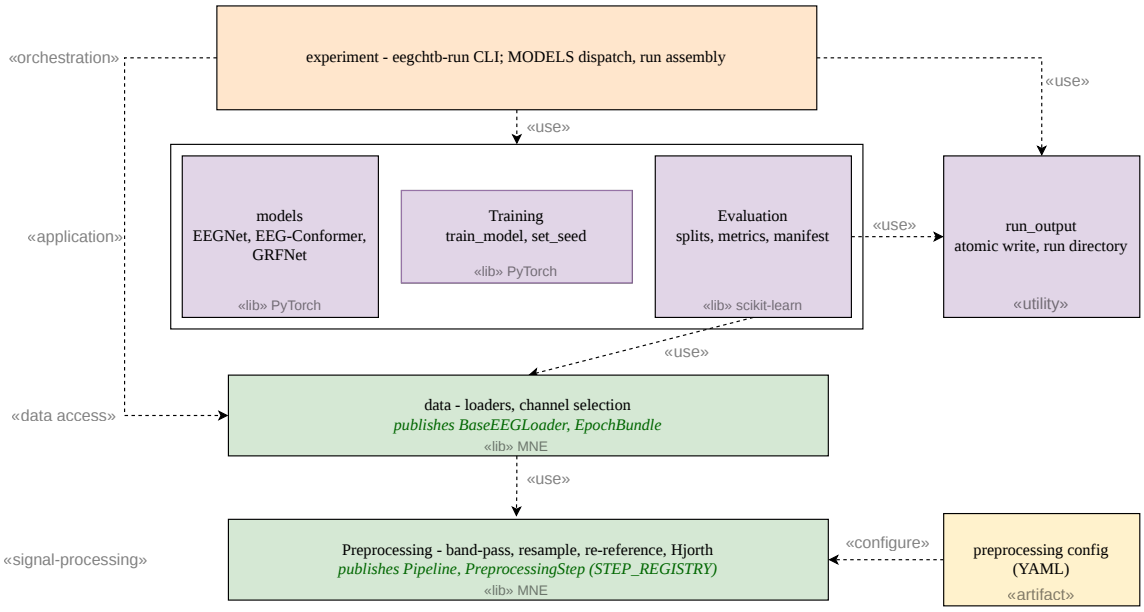


Figure 4.1: Layered package diagram of *eegchtb*

from the channel count and sampling rate, so any single architecture treats the montage density as an controlled experimental variable. Third, preprocessing is declared in YAML and applied to the continuous recording before epoching, this allows for not only the raw data from the BCI IV-2a and IV-2b but also the novel findings from the galea headset, no re-referencing is built into any model which enforces a separation of concerns.

4.2.1 Registry pattern and model subsystem

A key aspect of the framework is the registry pattern, based around the open/close principle. Preprocessing steps and dataset loaders have **PreprocessingStep** and **BaseEEGLoader** as abstract bases respectively, and a keyed registry where a step registers itself with a **@register_step** decorator into **STEP_REGISTRY** and loaders sit in **DATASET_LOADERS** so **build_step(name, params)** and **get_loader(name)** act as factories that fail without sufficient arguments.

This also allows for a new pipeline step or dataset to be added without touching any underlying code that interacts with it.

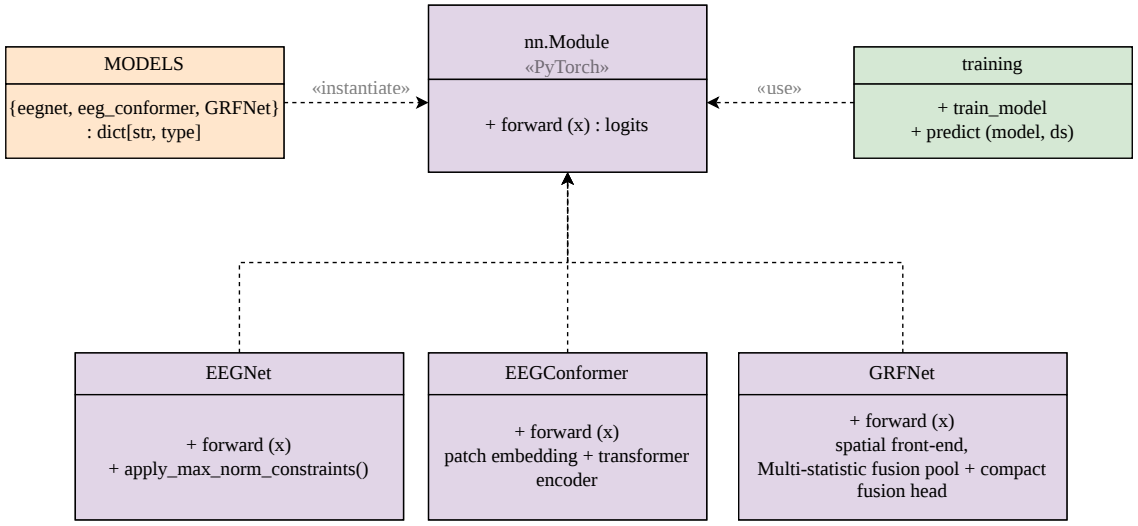


Figure 4.2: Class diagram of the model subsystem as implemented

4.2.2 The preprocessing subsystem

The preprocessing subsystem builds a **Pipeline** from **PipelineConfig** which reads from **YAML**. Each declared step is resolved through **build_step** against **STEP_REGISTRY** and applie in order of the continoue **mne.Raw**. Every step of preprocessing documented in 3.3 exposes **apply, params** and an optional **resolved_provenance** the Hjorth step overrides the last to record the neighbour graph and effective neighbour count that it resolves for the montage. The pipelines **resolved_config** reports every steps full parameter count which becomes the provenance record.

4.3 Model implementation and baseline validation

For validation of the three main models, validation scripts are provided to confirm that the model is implemented correctly by reproducing the accuracy the original authors reported using as close to the original model design and architecture as well as the exact protocol they used. Every model identifies its parameter count via **count_parameters** method in order to confirm the port matches the published parameter count. This creates blatant deviations when it comes to the benchmark grid as this has one shared, leakage safe methodology so that differences are attributable to the architecture.

4.3.1 EEGNet

EEGNet validation

EEGNet was validated as a reproduction rather than against a published baseline, EEG-Net was never evaluated on BCI Competition IV-2a, so every reported EEGNet figure is based on a later reimplementation known as EEG-TCNet [Ingolfsson et al., 2020]. For each of the nine subjects, the model is trained on the subjects training session (T) and evaluate on the held out sessions (E), under a *standard* preprocessing pipeline and on the full 22 channel montage for comparison with the published figure. The standardisation process and analysis window length are deliberately changed compared to the reference and recorded as controlled variables since fixing them through the *standard* pipeline for every model is the key point as to why EEGNet is comparable to the other architectures.

EEGNet training configuration

Training uses a fixed schedule against which the published figures are reported: 750 epochs over the whole training session with the Adam optimiser (learning rate 10^{-3}) and a batch size of 64, with no validation split or early stopping. The final epoch model is purposely the one to be evaluated, the max norm constraints are reapplied after every optimisation step. Each configuration is deterministic for a given seed and is repeated across fixed seeds to demonstrate run to run variability with every run scored by the chosen metrics.

EEGNet architecture implementation

EEGNet's port reproduces the original architecture, namely: A temporal convolution, a depth wise and a separable convolution. The original implementation being in Keras and translated to PyTorch for this project. The padding convention that adds an extra sample onto the right was implemented rather than PyTorch's default padding convention, Batch Norm momentum and epsilon are converted from Keras values to their PyTorch equivalent (0.99 to 0.01, 10^{-3}). The model has raw logits for pairing with **CrossEntropyLoss**. While Keras provides weight constraints that cap kernel norms, PyTorch does not do the same. Embedded constraints are contained within the optimiser, therefore the port has an

apply_max_norm_constraints method that the trainer calls after each optimiser step.

4.3.2 EEG-Conformer

EEG-conformer validation protocol

[Song et al., 2023] proposed EEG-Conformer and evaluated it under the competitions subject specific split, the validation script reproduces that protocol. For each subject, the model trains on session T and is tested on session E, using the references recipe (Adam, learning rate 2×10^{-4}), betas (0.5, 0.999), 2000 epochs, batch size 72, segmentation and recombination augmentation) and its default architecture. The reported checkpoint is leakage safe, best validation selection where the epoch with the lowest loss on a 15% slice is held out of session T.

EEG-Conformer implementation

There are two deviations from the reference: A per channel z-score rather than global z-scoring and seeded augmentation draws. EEG-Conformer's original implementation hard coded spatial convolution kernel size and attention head with for a 22 channel montage, meaning a reduction to four channels would dis-regulate the dimensions. The port therefore uses a dummy forward method to derive values from the montage it is operating on. Also, an unused classification token was removed with a correct key mask to avoid invalid results being produced.

4.3.3 GRFNet

GRFNet validation protocol

GRFNet is validated against MiniRocket developed by [Hwaidi and Ghanem, 2026b], due to the nature of it's influence on GRFNet, the validation script is specific to IV-2a under a pooled session protocol rather than a subject specific split. For each subject the two session T and E are combined and split into five folds as a trial, each held out fold is predicted once by a final epoch model using Adam, a learning rate of 10^{-3}, 800 epochs, batch

size 72 with segment recombine augmentation applied to the training folds. Euclidean alignment was used, this whitens each session by a reference covariance before pooling and is unsupervised.

GRFNet protocol deviations

GRFNets protocol contains the same session pooling but omits sub window segmentation because in the reference, leaks correlated within trial windows which could skew accuracy to present as higher than usual. This means that the gap in the reported implementation within this paper in comparison to the published results do not correlate to a build defect but rather that this projects protocol deliberately avoided results that may present as tailored to achieve a higher accuracy.

GRFNet architectural implementation

GRFNet is a novel development and therefore does not have one single reference architecture. MiniRocket/ROCKET, works by generating a large bank of convolutional kernels, freezing them and feeding their outputs through a linear classifier. The key reason for this approach is to resist overfitting specifically on small datasets due to the kernels not being trained. GRFNet implements this concept while tailoring it to the galea headset and BCI IV-2a/2b's characteristics, it contains a learnable spatial front end, a multi statistic fusion pooling stage and a gated linear head.

4.4 Practical implementation

Every loader returns an **EpochBundle**, a strict three dimensional array of **(trials, channels, times)** with its labels, sampling rate, channel names and provenance which checks against invariants and writes atomically.

4.4.1 Preprocessing configuration handling

The preprocessing pipeline is constructed from a YAML configuration where unknown step names fail at construction, the reusable deterministic trainer exposes optional per model hooks so a baseline can supply its own recipe, which has been demonstrated via the validation scripts.

4.4.2 Provenance record

Provenance is captured per run in a manifest, a hash of the sorted configuration (the fingerprint) is written and contains: the exact train, validation and test trials with automated leakage checks; the software, CUDA and device versions; the channels retained and used; the epoch window and the preprocessing specification. Outputs are written atomically via a temporary file and atomic replacement. Directories left on disk are those that are completed and substantiated.

4.4.3 Fingerprinting

The fingerprint gives each experiment a hash of the run inputs which allows identical setups to be recognised and every figure is traceable to the exact code, data and configuration that produced it.

The functionality behind this is the first sixteen hexadecimal characters of a SHA-256 digest taken over the run configuration with sorted keys, re-ordering the configuration will not change it as it is the same data. Every file is written to a temporary file and moved into place with **os.replace**, the run directory only persists if **.run_complete's** marker is written on clean exit. It deletes wholely on any exception so as not to have partial data runs cluttering the run directory.

4.4.4 Training loop

A single **train_model** function resolves the computation method (primarily CUDA, CPU backup), constructs the **DataLoader** with a seeded **torch.generator** so the mini batch

shuffle order is reproducible and optimises cross entropy with Adam under an optional cosine-annealing schedule. Its settings and outputs are concerned by to data classes, **TrainConfig** and **TrainResult**, giving a configuration and a structured record of the learning curves, best epoch and elapsed time. The early stopping, best checkpoint restore and max norm hook described above make the loop a single deterministic training strategy within the module.

4.4.5 System workflow

One single experiment resolves its configuration, loads the dataset, applies restrictions to galea channels, applies the chosen preprocessing pipeline to the continuous signal, cuts cue locked epochs into an **EpochBundle**, builds a seeded split, trains the selected model, predicts on a held-out session, scores with the key metrics and atomically writes results with a full manifest.

4.4.6 Error handling

Value objects validate their own invariants at construction, where an **EpochWindow** rejects a non positive window or an unrecognised unit, every **EpochBundle** checks on creation that its data array is three dimensional and that its label, channel name and time vectors match the array shape meaning an invalid trial container can't exist.

The same guards apply to models where constructors reject invalid hyper parameters, **forward** methods reject any input of the wrong value, the checks run at construction and at each layer boundary meaning a misconfiguration fails at an early point with sufficient error messages.

Comprehensive error handling is used in **__post_init__** across the whole codebase, it consistently raised **ValueErrors** to handle erroneous inputs.

4.4.7 Testing

The testing strategy employed is a verification of functionality and quality assurance aspect, for all key functionality there is a continuous integration check via these unit

tests. Quality is enforced by a pytest suite that is dependency and dataset gated allowing absences to result in a skipped test rather than a failed test. Every module carries forward, backwards, shape, param-count and input rejection tests. The trainer and runner carry determinism and atomicity tests with the determinism tests asserting bit identical preprocessing across reruns.

4.5 Challenges

4.5.1 Training collapse in the grid

The problem

The first pass through of the subject specific galea grid on BCI IV-2a across five seeds found that 14 of the 60 cells collapsed to one single class predictor. As mentioned in 3.9.2, Accuracy can stagnate on four class chance at 0.25 with Cohen's k at 0, this occurred majoritively on GRFNet under *standard*, *deep* and *maximum pipelines* with a single EEGNet cell (*standard*) affected as well.

The shared grid recipe started every architecture at the learning rate (10^{-3} from the first optimiser step, with early stopping patience at only 50 epochs. This combination resulted in the collapse, Z score standardisation rescales every input channel to unit variance so the randomly initialised first layer sees a larger input. This initial computation produced an oversized initial weight update which effectively snowballed the trajectory into instability and therefore caused the collapse.

The fix

The fix was implemented into the shared trainer, a 20 epoch linear learning rate 'warmed up' to the cosine decay which removes the oversized first update by gradually increasing the learning rate to full strength. Gradient norm clipping at 1.0 caps the size of any single update for the rest of training in order to prevent the spike recurring. The early stopping patience was raised from 50 to 250 epochs in order to allow the grid cell combinations to pass through rather than being stopped too early.

The outcome

Re-running the full 60 cell grid under the corrected recipe recovered all fourteen previous collapsed cells. GRFNet under *standard* preprocessing rose from the degenerate 0.250 ± 0.001 to 0.568 ± 0.120 which demonstrates a significant recovery. Across the whole grid, accuracy rose from 0.442 to 0.577 and no result remained a chance score of ≈ 0.25.

4.5.2 Inconsistent channel naming across datasets

The problem

BCI IV-2a and IV-2b identify electrode labels with different raw formats, this means that a channel such as C3 would present under different names when loaded, specifically for Hjorth Laplacian and current source density.

The fix

A normalisation step was added and applied at load time, it maps raw names onto one channel name scheme prior to preprocessing.

The outcome

Every step assumes a single naming convention regardless of the original dataset conventions.

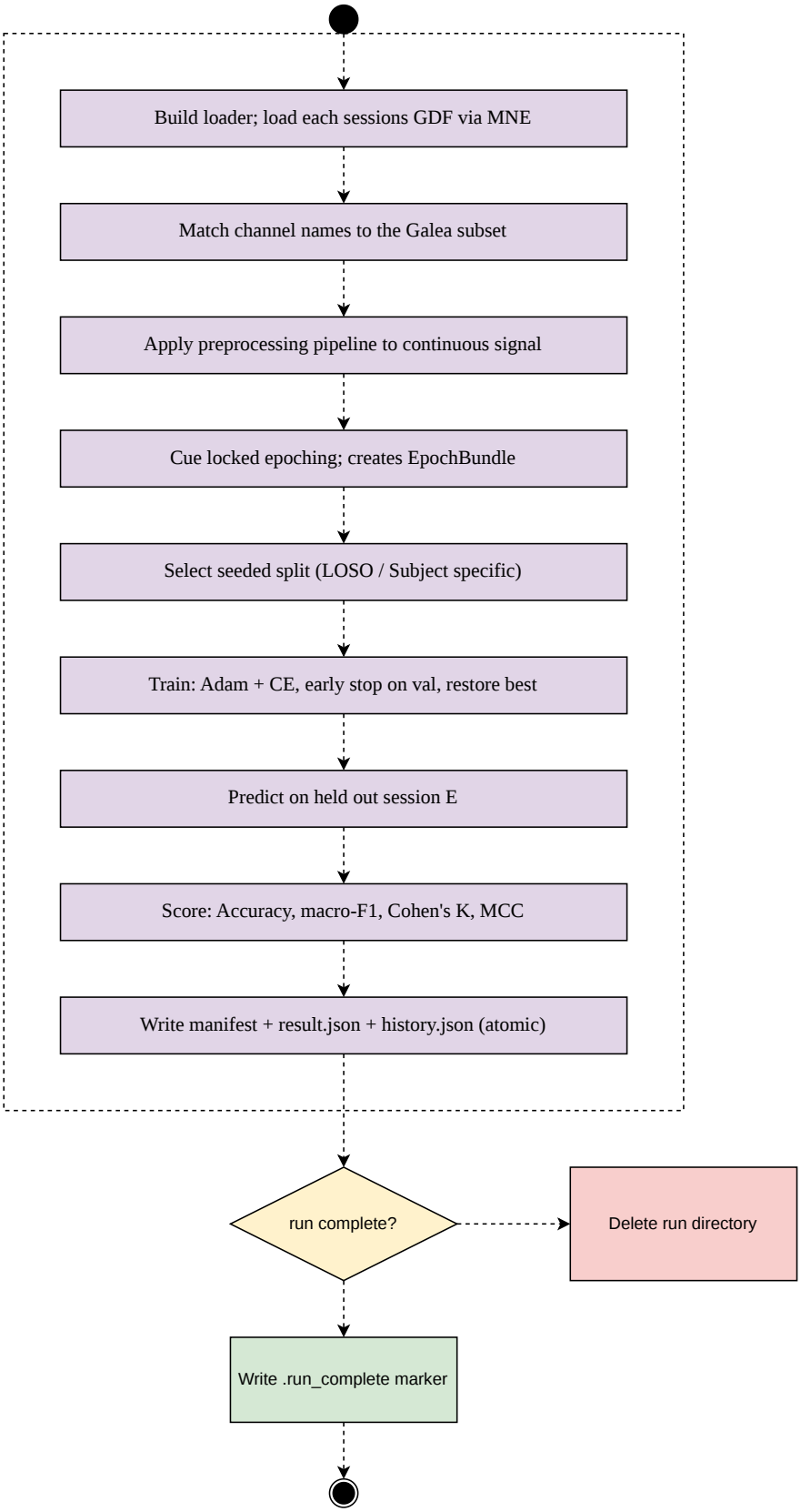


Figure 4.3: Activity diagram of a single experiment run

Chapter 5

Results / Discussion

This chapter reports on the empirical evidence produced by the validation scripts and the experimental benchmarking grid. The key metric to be reported in this section is accuracy due to the BCI Competition using accuracy as their primary scoring metric and is widely used throughout the field.

5.1 Baseline validation results

All **baseline** results below are on the *standard* preprocessing pipeline (a 4-40 Hz band-pass and per channel z-score) and the subject specific protocol. One model is trained per subject on session T and tested on session E. Each configuration is repeated across five random seeds and the reported figures are the mean and standard deviation across those seeds.

5.1.1 Full 22 channel montage validation results

Table 5.1 reports the five seed results.

5.1.2 Four channel Galea montage validation results

Table 5.2 reports the five seed results. It is clear that the reduction in electrodes to four distinctly lowers accuracy.

Model	Selection	Accuracy	Macro-F1	Cohen's κ	MCC
EEGNet	final-epoch, safe	0.709 ± 0.009	0.706	0.612	0.617
EEG-Conformer	best-test, optim.	0.807 ± 0.003	0.804	0.742	0.745
GRFNet	pooled, 5 fold	0.781 ± 0.002	0.780	0.708	0.711

Table 5.1: Five seed mean and standard deviation of the per-subject averaged metrics on the full 22 channel montage

Model	Selection	Accuracy	Macro-F1	Cohen's κ	MCC
EEGNet	final-epoch, safe	0.588 ± 0.007	0.581	0.451	0.459
EEG-Conformer	best-test, optim.	0.518 ± 0.008	0.472	0.358	0.374
GRFNet	pooled, 5 fold	0.653 ± 0.008	0.650	0.536	0.540

Table 5.2: Five seed mean and standard deviation of the per-subject averaged metrics on the four channel Galea montage

5.2 Grid results

The results below refer to the experimental grid, where three deep learning models are crossed with four preprocessing pipelines across 5 seeds, making 180 cells overall for the BCI IV-2a (120 cells) and IV-2b (60 cells)

5.2.1 IV-2a four channel galea grid results

Architecture	Minimal	Standard	Deep	Maximum	Mean
EEGNet	0.627	0.569	0.599	0.601	0.599
EEG-Conformer	0.558	0.522	0.594	0.589	0.566
GRFNet	0.562	0.557	0.574	0.571	0.566

Table 5.3: BCI IV-2a subject specific grid results on the four channel galea montage averaged over five seeds.

With regards to architecture, EEGNet leads with 0.559 mean accuracy, while the remaining models are tied at 0.566. This is important because GRFNet is able to achieve EEG-Conformers accuracy under this grids shared leakage safe protocol rather than under each architectures published validation, all while containing a fraction of the parameter count (GRFNet's 3655 to Conformers 789,572).

With regards to pipeline, Standard is the weakest with a 0.550 mean accuracy while the remaining are closer around ≈ 0.59. Although this presents as a weakness of standard, it

was most affected by the training instability that was corrected after the initial grid run and there may be smaller effects that haven't been ruled out.

Why are the means so closely correlated?

Within an architectures pipeline grid of 12 cells, accuracy varies across nine subjects with a mean standard deviation of 0.128, however across five random seeds within a cell, the spread is much tighter with around 0.027. This identifies that almost all variability comes from which subject is being decoded rather than which architecture is operated on.

5.2.2 IV-2b four channel galea grid results

Architecture	Minimal	Standard	Deep	Maximum	Mean
EEGNet	0.845	0.829	0.722	0.720	0.779
EEG-Conformer	0.816	0.796	0.741	0.736	0.772
GRFNet	0.769	0.781	0.684	0.685	0.730

Table 5.4: BCI IV-2b subject specific grid results on the four channel galea montage averaged over five seeds.

With regards to architecture, EEGNet and conformer are much closer together as to be expected, with GRFNet slightly. Comparing IV-2b to IV-2a by pipeline, it presents that minimal and standard modulation are now the strongest. IV-2b is a two class task with more concise electrodes chosen, this indicates that the heavier preprocessing pipelines may be over-modulating the signal and therefore removing key information.

5.2.3 IV-2a 22 channel grid results

Architecture	Minimal	Standard	Deep	Maximum	Mean
EEGNet	0.685	0.569	0.679	0.725	0.699
EEG-Conformer	0.655	0.611	0.703	0.665	0.659
GRFNet	0.635	0.639	0.706	0.699	0.670

Table 5.5: BCI IV-2a subject specific grid results on the 22 channel montage averaged over five seeds.

With regards to architecture, GRFNet exceeds EEG-Conformer at only 3655 parameters

to 789,572 respectively. The mean of means on IV-2a's 22 channels reports a gain of 0.099 left in the 18 channels which demonstrates the importance of the sensorimotor cortex for task classification. It also shows that the extra parameter capacity does not function as a better decoder.

5.3 Statistical analysis

5.3.1 Above chance

All 36 conditions decode significantly above their four class chance level of 0.25. The pooled binomial test on each conditions test trials produces Holm-adjusted $p$values below 10^{-300}. Every condition clears one sided Wilcoxon signed-rank test at its resolution floor (p = 0.0020 to 0.0059). This therefore reinforces the initial assumption that four electrodes carry decodable motor imagery information across every architecture and preprocessing modulation.

5.3.2 Architecture

The Friedman omnibus is non significant in all four 22 channel IV-2a cells ($p = 0.121$ to 0.641, $W = 0.05$ to 0.23) and significant in all eight sparse montage cells ($p = 0.0018$ to 0.0446, $W = 0.35$ to 0.70). Architecture is interchangeable when the montage is rich and separable when it is sparse, this finding supports the projects sparse nature. 8 of 36 pairwise comparisons survived Holm correction as shown in table 5.6. EEGNet is the strongest model overall beating both models considerably.

5.3.3 Preprocessing

The Friedman omnibus is significant in five of nine cells, the direction of the effect reverses between datasets. For IV-2a, heavier preprocessing aids the accuracy whereas on IV-2b it hinders 0.12 accuracy for EEGNet. Table 5.7 shows that seven pairwise contrasts survive Holm correction, a possible explanation is the signal budget, IV-2b's three bipolar

Experiment	Pipeline	Comparison	Med. diff	p(Holm)
2a / galea	minimal	Conf vs EEGNet	-0.056	0.012
2a / galea	standard	Conf vs EEGNet	-0.051	0.012
2b / galea	minimal	Conf vs EEGNet	-0.029	0.012
2b / galea	deep	Conf vs GRFNet	+0.047	0.039
2b / galea	deep	EEGNet vs GRFNet	+0.026	0.012
2b / galea	maximum	Conf vs GRFNet	+0.036	0.039
2b / galea	maximum	EEGNet vs GRFNet	+0.026	0.023
2b / galea	standard	EEGNet vs GRFNet	+0.043	0.012

Table 5.6: Pairwise architecture comparisons through Holm-Bonferroni correction

channels is subject to miniscule redundancy meaning artefact handling can remove signal along with noise whereas IV-2a is not subject to this.

Experiment	Model	Comparison	Med. diff	p(Holm)
2a / all	GRFNet	deep vs minimal	+0.022	0.039
2a / all	GRFNet	deep vs standard	+0.045	0.023
2a / galea	Conformer	deep vs standard	+0.070	0.047
2b / galea	EEGNet	deep vs minimal	-0.093	0.039
2b / galea	EEGNet	deep vs standard	-0.119	0.039
2b / galea	EEGNet	max vs minimal	-0.086	0.023
2b / galea	EEGNet	max vs standard	0.110	0.039

Table 5.7: Pairwise preprocessing comparisons through Holm-Bonferroni correction

5.3.4 Montage cost

Restricting IV-2a from 22 electrodes to four presents significance in 9 of the 12 cells that were tested (three models by four preprocessing pipelines). The three non significant cells fail on statistical power. Holm correction over twelve tests requires a raw p at or below 0.0042, while the signed-rank test bottoms out at 0.0039. The loss is positive in all 12 cells and every BCa interval excludes zero.

5.3.5 Seed stability

The within subject standard deviation across five seeds has a median of 0.033 accuracy, against a median absolute pairwise model difference of 0.026. Differences in the size of the noise floor is only detectable through seed averaging which identified that it is larger

than the effect being tested.

5.4 Galea recordings transfer analysis

This section tests whether any of the decoders transfer to consumer hardware after being trained on IV-2a. Three participants recorded on OpenBCI's Galea headset under a similar motor imagery paradigm and scored by every snapshot in the trained grid on a transfer from the benchmark grid to the Galea recordings.

Source	Cells	Mean	Range	Chance
IV-2a to Galea	36	0.253	0.190 - 0.308	0.250
IV-2b to Galea	36	0.499	0.437 - 0.583	0.500

Table 5.8: Transfer from the benchmark grid to Galea recordings, Mean accuracy over 45 snapshots per cell.

5.4.1 Transfer results

Neither transfer exceeds chance, Macro-F1 spans 0.11 to 0.25 in four class and 0.34 to 0.53 in binary which confirms the models produce a distributed prediction. Using the best cell per recording in IV-2b and testing it against chance gives 11 of 20 for A, 8 of 14 for D and 7 if 14 for J. This binomial test is used because each trial is an independent binary outcome against a known chance which is optimal for a one sample proportion test.

5.4.2 Cross subject control

Class	Within	Cross	Galea	Chance
2a, 4 class, 4 ch	0.577	0.317	0.253	0.250
2b, 2 class, 3 ch	0.760	0.618	0.499	0.500

Table 5.9: Cross-subject control from within subject, cross subject and galea transfer accuracy

Where cross subject transfer exceeds chance, the same models fall to chance on Galea specifically. Considering cross subject variance has survived even if by a small margin, it indicates that the gap from 0.618 to 0.499 is specific to the galea recordings.

5.4.3 Noise impedence

Source	50 Hz / 4-40 Hz	70-100 Hz floor	4-40 Hz power
Galea A	5.75	0.062	1.39
Galea D	91.33	0.310	2.53
Galea J	13.10	0.151	1.19
2a S01	0.04	0.164	1.24
2a S02	0.18	1.419	2.85
2a S03	0.07	0.434	1.77

Table 5.10: Noise characteristics on sensorimotor channels (C3, Cz, C4) - Relative line noise, noise floor and signal power

Relative line noise is significantly higher on Galea, this is a key indicator of high electrode-scalp impedance, poorly coupled electrodes reject common interference poorly. The band pass removes 50 Hz before classification so features aren't directly corrupted but quality is hindered.

This is also partly due to the fact that, when comparing wet saline Ag/AgCl electrodes to dry polymer pillar electrodes, [Di Flumeri et al., 2019, Faadhilah Afif et al., 2020] found that impedance is typically higher with dry electrodes. Wet Ag/AgCl electrodes benefit from electrolyte interface where salie provides a conduction pathway between the electrode and scalp which improves the charge transfer [Lee et al., 2022]. Dry polymer pillar electrodes rely solely on direct contact with the scalp which makes the interface struggle when contact is not constant and applied at pressure. Although pillar geometries increase the effective contact area, dry electrodes commonly suffer with impedances ranging from 10-100 kOhms which has been demonstrated by the unique galea recordings collected.

5.5 Further discussion

The results identify that the project hypothesis that reported accuracy advantages of heavy architectures would not survive on sparse consumer grade montages is supported by thorough grid evidence and that smaller compact decoders can perform just as well across a wide array of variables has also been proven true.

The first key finding is that Table 5.5 identifies that a transformers capacity does not

present any advantages under controlled and shared protocols. The validation script results present higher accuracies due to the tailored evaluation and preprocessing which is specific to the model and is therefore not generalisable.

The second key finding is that GRFNet achieves its design goal of compact efficient decoding with limited parameters and tailoring to the Galea headset. It has fewer parameters than EEGNet (3655 compared to 3932) and matches EEG-Conformers mean accuracy across the grid on the four channel Galea montage (5.3). On the full 22 channel montage GRFNet exceeds EEG-Conformer (0.670 to 0.659), this indicates that ROCKET inspired fixed feature cores and multi-statistic fusion pooling are utilised for decodable motor imagery signal without large convolutional layers as well as supporting a low over-fitting recipe. While there is still room for improvement, the cross seed variance expresses that differences are down to the architecture, which was not tuned via hyper-parameters or optimised in any way before its deployment.

The third key finding is that the pipeline ranking is dataset dependant rather than montage dependant, IV-2a grids agree that high modulation intensities are optimal whereas IV-2b considers the less modulating pipelines to be more effective which suggests the pipeline effect interacts with the classification task rather than electrode count.

These findings identify a narrative that the transformer capacity is present for dense research-grade montages under optimistic protocols but does not blatantly identify architectural advantages when removing the favourable variables. GRFNet has demonstrated that a fixed kernel feature bank with 3655 parameters can capture clear temporal structures of motor imagery EEG data without learning spatial relationships. The pipeline reversal indicates that there isn't an individual preprocessing pipeline across all motor imagery publications which leads to the conclusion that systems must be validated and tailored to its own data rather than inheriting protocols.

With regards to the hardware, Galea's electrodes are a conductive pillar polymer and do not have the key features of dry electrodes which makes operating around a participants hair significantly more reliable, pin and multi spike crown geometries are optimal for dry electrode designs because it parts hair and holds it aside. These designs report impedances

of 7 to 25 kOhms compared to wet Ag/Ag-Cl electrode impedances of 10 kOhms. The measured line noise ratios of 5.75 to 91.3 against IV-2a's 0.04 to 0.18 identify that the recordings have a high impedance.

5.6 Result limitations

5.6.1 Subject metadata

One limitation is the subject metadata, in which neither official dataset publishes individual participant details, IV-2a documentation [Brunner et al., 2008] reports only a subject count of anonymous ID's with no age, gender, handedness, or health status disclosed for any subject, IV-2b documentation [Leeb et al., 2008] states that participants were right handed and had normal vision. This dissertation can't condition its per subject variance on demographics or provide an explanation for 'hard subjects' and their lower decoding accuracy. It can be stated that some of the 'hard subjects' may be part of a third of participants who are 'BCI illiterate' in which users produce signals that are difficult to decode [Vidaurre and Blankertz, 2010] but this can not be confirmed.

5.6.2 Electrode approximation

Given that the four channel subject, named the Galea channel montage in this paper, is only an approximation of the overlapping electrodes and not directly the Galea headset itself, external validity has not practically been demonstrated and remains to be done in the future.

5.6.3 Trial counts

Trial counts for the Galea recordings are at 7 for D and J and 10 for A, this can indicate that the results are simply underpowered, given that IV-2a trials were at 288 per session. As well as this, ocular contamination was uncorrected, cue order was cyclic and electrode positions were not verified. All of these extraneous variables can lead to null results.

Chapter 6

Conclusion and Future Work

6.1 Contributions and Reflections

6.1.1 Contributions

Novel standardised benchmarking framework utilising Galea The key contribution of this project is the benchmarking framework. It is deterministic and controlled where every architecture passes through one shared loader, split, training, scoring and provenance path. There has not previously been a comparable study in which deep learning architectures have been compared not only on differing modulation intensities of the preprocessing pipelines they use, but also the effects that a sparse consumer grade montage such as Galea has on key metrics that are widely used. Key features such as montage adaptive model implementations, per run fingerprinting and atomic output alows for the extension of this benchmark without modifying internal logic that handles the paths.

Novel architecture: GRFNet A second contribution is GRFNet, a compact, ROCKET inspired decoder. It contains multi statistic fusion pooling which utilises PPV, log variance band power and a mean and concatenates and softmax weights these streams so the classifier analyses the statistics separately. It matches EEG-Conformers grid mean on the Galea montage and exceeds it on the full 22 channel montage on standardised preprocessing pipelines at a significantly smaller parameter count.

6.1.2 Reflections

What went well

Consistent development methodology The development methodology was applied consistently and managed to catch a large training collapse problem (4.5.1). The per cell fingerprinting made the collapse traceable to specific cells which helped identify the possible reasons rather than an overall accuracy drop that may have lead me down a more wholistic path. The four metrics reported (accuracy, Macro F1, Cohen's k and MCC) demonstrated the reason they are used throughout this problem, Cohen's k exposed the class collapse expressed as $k \approx 0$ where accuracy would have only shown that the runs were achieving a class chance of 0.25.

PyTorch Porting The decision to use PyTorch ports rather than brain-decode implementations gave the project more control over the model interface allowed modification in spaces such as EEG-Conformer in 4.3.2, where deviations were necessary for functionality within the grid.

Improvements

The initial shared training recipe was adopted without a smoke test for stability. These tests, if implemented would have identified things such as the gradient warm-up requirements before the full grid was due to be run. Similarly, a smoke test of the estimated time to run the validation script on a transformer architecture with a learning rate of 2×10^{-4} on 2000 epochs may have been beneficial given that it was by far the longest runs across this project.

6.2 Legal, Social, Ethical and Professional Issues

6.2.1 Legal

EEG recordings are considered personal data capable of identifying individuals and therefore are identified within the scope of the UK General Data Protection Regulation pre-

sented by [DPA, 2018].

BCI competition dataset collection The BCI IV-2a [Brunner et al., 2008] and IV-2b [Leeb et al., 2008] datasets were collected under their own institutional approvals and released for research use as an open source dataset, the requirements dictate that appropriate references must be applied and that redistribution for commercial profits are restricted. It is also the contention that if the datasets were to be stored in a closed source repository, it would constitute redistribution and therefore be restricted under the publishers guidelines.
No attempt is made to identify the participants, the datasets are acknowledged and are not included except for relevant links to their download to reproduce the benchmark.

Galea recording collection The Galea recordings collected for the transfer study involve new human participants, informed consent was obtained from each participant, recordings are stored within the repository as the scope of this project is confined to the university, specifically for research and testing purposes, under which the recordings are ethically permitted. Reference implementations are referenced via code blocks as well as cited in this paper.

Intellectual property Reference implementations are adapted from open-source projects and are sufficiently cited, the project's code if published would become open source and would need to gain ethical approval and a relevant license to dictate redistribution and usage.

6.2.2 Social

BCI contributions Motor imagery BCI's purpose is to restore motor functionality to users who are impaired, research that clarifies which architectures are able to be applied across a range of functionality requirements is beneficial for society. Similarly, clarifying the extent to which a consumer grade, relatively cheap headset with EEG capabilities can perform and be extrapolated into more popular avenues of BCI analysis is also a

contribution to society which may provide more accessible BCI development.

Dual use concerns There is a concern of dual use in which motor imagery decoding pipelines can unjustly be applied to manage mental workload in the workplace. UNESCO's recommendation on the ethics of neurotechnology identifies that surveillance applications and un-consensual cognitive monitoring is a risk that must be cautioned against [UNESCO, 2025]. This project mitigates the risk by establishing that the data is specifically only for testing/ research purposes.

Evaluation inclusivity Cross subject generalisation can present inclusivity issues where long sessions may exclude those who are functionally unable, Reporting cross subject accuracy alongside subject specific results provides a more generalisable study of real world deployment and is a focus for future work.

6.2.3 Ethical

Galea recordings approval The primary concern is the collection of neurological data from three participants for the Galea recordings. EEG signals can reveal cognitive states beyond the specific task being studied, collection is limited to the minimal trials required for evaluation and recordings are not publicly available. Ethical approval is not required because the data is simply used for testing rather than any published work.

Generative Artifical Intelligence Generative AI use is limited to collating notes regarding the overall topic of EEG, Machine and Deep learning. All information is fact checked against reliable sources and no content has been copy and pasted.

Environmental sustainability Training deep neural networks consumes more electrical energy than standard programming tasks due to the benchmarking grid's large amount of cells. To minimise waste, re-runs are avoided where possible and cloud computing is limited to an emergency situation only with its energy consumption estimated and logged.

6.2.4 Professional

This project is conducted in accordance with the [BCS, The Chartered Institute for IT, 2022]. It states a duty to act within a professional competence, this ensured that a plan was formed to validate that EEGNet's baseline was achievable before continuing to more difficult work. The duty of integrity specifically states that all findings must be reported honestly, including null or unfavourable results, in this circumstance, the Galea transfer study's null result is reported in full and accompanied by explanations and possible reasons as to the variables that may have caused it. The code presents a duty to the public interest which aligns with this projects semantic purpose of identifying honest accuracies of published architectures, honest reporting and identifying constraints with a consumer grade headset.

6.3 Synoptic Reflections

My overall ability to conduct independent analysis, research and development has improved greatly.

Software engineering and system design Building the **eegchtb** framework required developmental consistency to learn as I built as well as applying my programming and design knowledge throughout. There was constant consideration of a separation of concerns, dependency inversion, test first principles and contract principles, where the experience of diagnosing and fixing system wide failures reinforced practical problem solving and the value of the aforementioned principles.

Machine learning and signal processing Implementing three architectures through PyTorch and porting from Keras/Tensor-flow required a working understanding of each models functionality beyond the library functions. Many developmental tasks improved my problem solving skills, critical analysis of the task, and overall programming ability as my perception of coding is majoritively searching up key library functions, understanding them and effectively implementing them.

Research and critical analysis My literature review and project methodology required in depth analysis of a variety of papers, articles and other sources to determine what was relevant enough to be included within the paper. My researching efficiency noticeably improved as the amount of content I considered using also increased. My critical analysis skills improved where, writing up the results with limitations and doing seed noise diagnostics at the same time, rather than falsely claiming from descriptive statistics, undertook statistical analysis across the board as a key feature of published work.

6.4 Future Work

One main idea of future work is to benchmark the full 10 channel electrodes from the Galea headset onto Galea recordings which would test GRFNet's geometry aware design on the electrode count it was ideally designed for.

A second idea is cross subject evaluation which under the time constraints of this project was not able to be completed. It would test whether the architecture and pipeline rankings observed under subject specific training is able to be generalised to unseen subjects, this is not sufficiently demonstrated on unseen Galea recorded subjects due to the null results.

A final potential idea with regards to the Galea headset would be to record significantly more trials, with a more standardised and controlled experimental methodology, then train the models using the same splits on the Galea data and run the entire grid to see the results, it may be the case that both subject-specific and cross subject evaluations could be used to identify the extent to which training is generalisable to unseen subjects. This remains a large problem in EEG Deep learning and therefore always has more research to be undertaken regarding it.

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Appendix A

Galea recording methodology

The methodology sets out the Galea markers in relation to the motor imagery class, as well as documents the specific imagined movements.
(n) refers to the Galea marker that is inputted via the Galea GUI

```
2 Minutes eyes open - (2)

1 Minute eyes closed - (3)

1 Minute eye movement - (4)


Imagined movements

Tongue - Pointing forward -      [4 seconds]    (5)

Left hand - slow open and closed [4 seconds]    (6)

Right hand - clenching           [4 seconds]    (7)

Both feet - Rotating both ankles [4 seconds]    (8)
```

Appendix B

Preprocessing pipeline configuration

The four preprocessing pipelines are declared in a YAML configuration file by the **Pipeline** class.

```
# minimal.yaml
name: minimal
steps:
  - name: bandpass
    params:
      l_freq: 0.5
      h_freq: 100.0
```